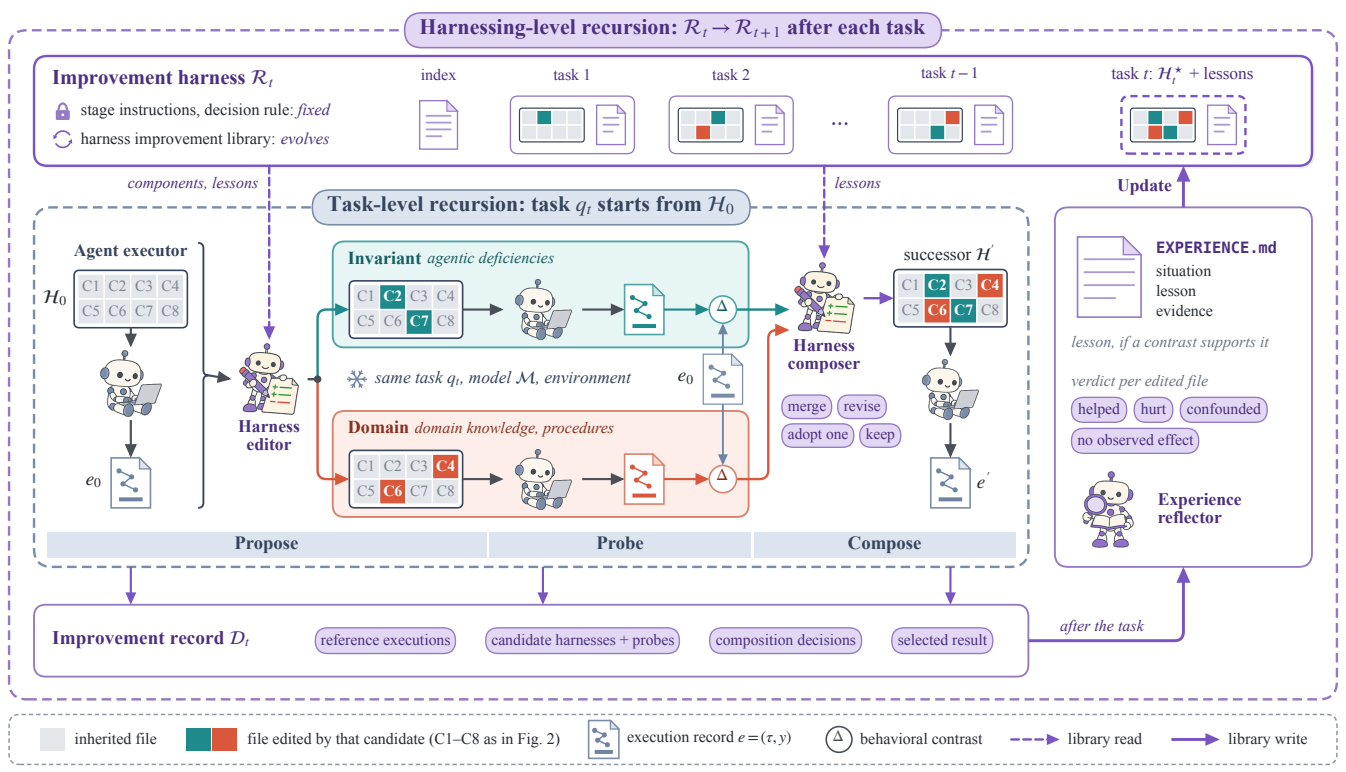

Harnessing-level recursion: $\mathcal{R}_t \to \mathcal{R}_{t+1}$ after each task
Improvement harness $\mathcal{R}_t$
stage instructions, decision rule: fixed
harness improvement library: evolves
index
task 1
task 2
task $t-1$
task t: $\mathcal{H}_t^\star$ + lessons
components, lessons
lessons
Update
Task-level recursion: task q_t starts from $\mathcal{H}_0$
Agent executor
$\mathcal{H}_0$
C1 C2 C3 C4
C5 C6 C7 C8
e_0
Harness editor
Invariant agentic deficiencies
same task q_t, model $\mathcal{M}$, environment
Domain domain knowledge, procedures
Harness composer
merge
revise
adopt one
keep
successor $\mathcal{H}'$
e'
EXPERIENCE.md
situation
lesson
evidence
lesson, if a contrast supports it
verdict per edited file
helped
hurt
confounded
no observed effect
Experience reflector
Propose
Probe
Compose
Improvement record $\mathcal{D}_t$
reference executions
candidate harnesses + probes
composition decisions
selected result
after the task
inherited file
file edited by that candidate (C1–C8 as in Fig. 2)
execution record $e = (\tau, y)$
behavioral contrast
library read
library write